SOLAR LIGHTS LITE / PRINTABLE SHEET

Controller bench sheet

Flash the advanced build, prove the controller, then measure the LED string. Gate 1 of docs/10-commissioning.md.

> **Before you start**
> No cells connected for any of this. Use a current-limited bench supply, 200 mA limit, for VCC.
> Step 2 needs the regulator and power LED removed (Gate 0) and the bench supply as the only power source.
> Do not connect the panel or the charger. R3 sits at the board end, not out at the string.
> Duty does not limit current: only R3 and the pin do. Verify at 4.20 V and 100% duty.

1 Flash the advanced build

Fuses are set once per board (internal 8 MHz, BOD 2.7 V, no bootloader). Routine flashing never writes them.

Disconnect cells, panel, UART and LED string. Power from one verified target-compatible ISP source only.

Action	Command / pass result
Open firmware folder	cd firmware/controller
Check the target	avrdude -c usbasp -p m328p -v -> signature 1E 95 0F
	1E 95 16 = ATmega328PB: use -p m328pb, never -F
Compile advanced build	pio run -e advanced -> SUCCESS
Upload and verify	standalone current avrdude; never pio -t upload
First power-up check	USB-UART at 3.3 V into serial-header VCC -> banner; vdd within 0.05 V of meter

Disconnect all ISP wires first. Then remove the regulator and power LED (guide 04 step 5) and repeat the check.
After that, USB-UART is monitoring-only: target TX -> adapter RX, GND -> GND; adapter VCC, TX, DTR, CTS open.
Keep the adapter on USB while it is wired: unplugged, it loads TX and garbles the telemetry. Unwire it first.

Tool setup, fuses, Arduino-as-ISP fallback and troubleshooting: docs/08-firmware.md

2 Check the controller on a current-limited bench supply

Read the diagnostics through Tier 1 telemetry or a USB-UART adapter (RX and GND only) at 9600 baud.

Check	Expect
Startup after reset	Banner with version and build date, field guide, then diagnostics
VCC at 3.80 V	vdd within 0.05 V of the meter (config.h BANDGAP_V = 1.10 x Vmeter / Vreported)
D9 PWM frequency	1.9 to 2.0 kHz, confirming the 8 MHz clock
Pull D2 to GND	cause=BUTTON_TEST, output=100%, button timer counts down
Cover / uncover LDR	DUSK / DAWN confirmation reaches 300 s before mode changes

3 LED string current with R3 = 56 ohm

Meter in series with the string positive, lights at 100% duty (press D2 or cover the LDR). Expect = model:

Supply	Expect	Measured	R3 temp	Notes
3.40 V	10 mA	_____	_____	_____
3.70 V	13 mA	_____	_____	_____
4.20 V	19 mA	_____	_____	_____

> **Decision rule**
> Over 20 mA at 4.20 V: fit 68 ohm and repeat the three readings (expect about 17 / 12 / 9 mA).
> Too bright or want longer run time: lower NIGHT_DUTY_PCT in config.h. Too dim: this design is at its limit; a brighter string needs a driver stage (docs/12-going-further.md).

Also confirm: R3 and the D9 pin stay at ambient; the string is dark with M2 unplugged and during reset.

For deployment, build pio run -e production (or standalone) and flash its .hex the same way.